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(Affiliated to the University of Mumbai)
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DEPARTMENT OF COMPUTER ENGINEERING

MINI-PROJECT REPORT ON

GREENCITY – CLOUD-BASED SMART WASTE MANAGEMENT AND COMPLAINT TRACKING SYSTEM

Submitted in Partial Fulfillment of the Requirements for the Degree of
Bachelor of Engineering in Computer Engineering (Semester IV / Cloud Computing)
Academic Year: 2026–2027

SUBMITTED BY:
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UNDER THE GUIDANCE OF:
Prof. Sangeeta Parshionikar
Assistant Professor, Department of Computer Engineering

2. CERTIFICATE OF APPROVAL

This is to certify that the mini-project entitled "GreenCity – Cloud-Based Smart Waste Management and Complaint Tracking System" is a bona fide work carried out by Sakshi (Roll No. 10895) in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Engineering at Fr. Conceicao Rodrigues College of Engineering, affiliated to the University of Mumbai, during the academic year 2026–2027.

This individual mini-project has been completed under my supervision and guidance and satisfies the academic standards and evaluation criteria prescribed for the Cloud Computing curriculum.

_____	_____
Prof. Sangeeta Parshionikar	**Head of Department**
Project Guide, Dept. of Computer Engg.	Dept. of Computer Engg.
_____	_____
Internal Examiner	**External Examiner**
Date: _____	Place: Mumbai

3. STUDENT DECLARATION

I hereby declare that the mini-project entitled "GreenCity – Cloud-Based Smart Waste Management and Complaint Tracking System" submitted by me to the Department of Computer Engineering, Fr. Conceicao Rodrigues College of Engineering, Bandra, Mumbai, in partial fulfillment of the requirements for the degree of Bachelor of Engineering in Computer Engineering, is an authentic record of my own individual work carried out under the supervision of Prof. Sangeeta Parshionikar.

I further declare that this project is my original work and has not formed the basis for the award of any degree, diploma, associate-ship, fellowship, or similar title to any other candidate at any university or institution.

_____

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4. ACKNOWLEDGEMENT

I express my deepest gratitude to my project guide, Prof. Sangeeta Parshionikar, for her invaluable guidance and encouragement throughout the development of my Cloud Computing mini-project. I also express my sincere thanks to the Head of Department and the Principal of Fr. Conceicao Rodrigues College of Engineering for providing the necessary facilities.

5. ABSTRACT

GreenCity is a cloud-architected smart waste management and complaint tracking platform engineered to modernize municipal solid waste workflows. Built on a modular 3-tier architecture, GreenCity integrates: (1) A modern React/Vite single-page application with responsive dashboards, interactive GIS bin mapping via Leaflet, and statistical visualization via Recharts; (2) A scalable Node.js/Express REST backend implementing JWT authentication, role-based access control (RBAC), and image processing with cryptographic magic-byte validation; (3) An enterprise-grade AWS Cloud Infrastructure featuring a custom Virtual Private Cloud (VPC), multi-AZ public and private subnets, Amazon EC2 compute instances, Amazon RDS MySQL in private isolation, Amazon S3 for secure photo object storage with Pre-signed URLs, AWS Systems Manager (SSM) Parameter Store for zero-exposure credential injection, Amazon CloudWatch for log streaming and alarm monitoring, and AWS Budgets for cost predictability.

The system was verified through an automated end-to-end test suite containing 24 test cases spanning all functional modules, achieving a verified 100% pass rate (24/24 tests passed). GreenCity directly advances United Nations Sustainable Development Goals 11 and 12.

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7. INTRODUCTION & 8. PROBLEM STATEMENT

7. Introduction

Municipal solid waste management represents one of the most critical urban challenges of modern cities. Rapid urbanization and high population densities strain civic collection infrastructure, leading to overflowing bins, environmental hazards, and unaddressed citizen grievances. GreenCity provides an end-to-end cloud platform that automates civic waste monitoring and complaint redressal.

8. Problem Statement

1. **Operational Blindspots:** Municipalities lack real-time visibility into bin capacities and collection progress.
2. **Fragmented Feedback:** Citizens lack a verifiable channel to report overflowing bins with photo evidence.
3. **Fleet Allocation Deficits:** Vehicles operate on rigid manual schedules rather than demand-based collection.
4. **Legacy Server Fragility:** Municipal servers lack automated backups, network segmentation, and observability.

9. OBJECTIVES & 10. PROJECT SCOPE

9. Objectives of GreenCity

Objective 1: Engineer an intuitive React SPA for complaint submission, GIS mapping, and fleet tracking.

Objective 2: Implement an asynchronous Node.js REST API with JWT security and image validation.

Objective 3: Architect a Well-Architected AWS 3-tier cloud topology (VPC, EC2, RDS MySQL, S3, SSM, CloudWatch).

Objective 4: Execute automated testing covering 100% of functional modules.

10. Project Scope

The project encompasses urban civic wards, providing citizens with digital complaint tracking and municipal supervisors with a centralized dashboard for fleet, pickups, and GIS mapping.

11. EXISTING SYSTEM & 12. PROPOSED SYSTEM

11. Existing System

Current municipal systems rely on manual register logs, phone helplines, and fixed daily routes. This leads to slow complaint resolution, fuel wastage, and zero operational transparency.

12. Proposed System

1. **Verifiable Complaints:** Citizens report waste with photo evidence; supervisors track lifecycle states.
2. **Interactive GIS Mapping:** Leaflet map displays city-wide waste containers with fill-status indicators.
3. **Fleet Scheduling:** Supervisors dispatch compactor and tipper trucks based on actual collection demands.
4. **3-Tier Cloud Security:** RDS MySQL isolated in private subnets, SSM secrets management, and CloudWatch alarms.

13. SYSTEM REQUIREMENTS & 14. TECHNOLOGIES USED

13. System Requirements

Hardware: Development PC (Core i5/i7, 8-16 GB RAM); Cloud compute (EC2 t3.micro, RDS db.t3.micro).
Software: Node.js (v20/v24 LTS), MySQL 8.0, React 19, Vite 6, Leaflet 1.9, Recharts 2.15, AWS CLI.

14. Technologies Used

Frontend: React 19, Vite, Leaflet GIS, Recharts, Lucide Icons, CSS Modules

Backend: Node.js, Express.js 5, Multer, Bcrypt, JSON Web Tokens (JWT)

Database: MySQL 8.0 Relational Engine with mysql2/promise connection pool

15. CLOUD CONCEPTS & 16. CLOUD ARCHITECTURE

15. Cloud Computing Concepts

In GreenCity, I implemented key cloud principles: IaaS for virtual compute (EC2) and networking (VPC); PaaS for relational database management (RDS); Object Storage for complaint photos (S3); Decoupled 3-tier architecture; CloudWatch observability; and AWS Budgets cost governance.

16. Cloud Architecture

VPC CIDR 10.0.0.0/16 with 2 public and 2 private subnets across Multi-AZ (us-east-1a, us-east-1b). Internet Gateway routes public traffic. RDS MySQL is placed in private subnets with access restricted strictly to the EC2 App Security Group on port 3306.

17. SYSTEM ARCHITECTURE & 18. DATA FLOW DIAGRAM

17. 3-Tier Architecture

Tier 1 (Presentation): React SPA running in browser, connecting via HTTP 80/5000.
Tier 2 (Application): Amazon EC2 instance in Public Subnet running Node.js REST API with Nginx.
Tier 3 (Database): Amazon RDS MySQL in isolated Private Subnets with restricted security group.

18. Data Flow Overview

Citizen -> Submit Complaint & Photo -> Backend verifies magic bytes -> Stores in S3/DB -> Supervisor reviews on Dashboard -> Dispatches Fleet Truck -> Clears Bin -> Marks Resolved.

19. DATABASE DESIGN & ER DIAGRAM

19. Database Entities

users: id, name, email (UQ), password (bcrypt hash), role, status, created_at

complaints: id, name, location, category, description, image_key, status, created_at, updated_at

vehicles: id, vehicle_number (UQ), vehicle_type, driver_name, driver_phone, capacity, status

waste_collections: id, location, waste_type, quantity, collection_date, vehicle_number, status

collection_locations: id, location_name, address, area, latitude, longitude, bin_capacity, status

20. FRONTEND & 21. BACKEND IMPLEMENTATION

20. Frontend Implementation

Built with React 19 and Vite. Implements src/config.js for dynamic API routing, Leaflet GIS mapping with custom color-coded status markers, Recharts visual analytics, and client-side form validation.

21. Backend Implementation

Built with Node.js and Express. Implements mysql2/promise connection pool, multer in-memory file streaming with magic-byte validation, bcrypt password hashing, signed JWT token validation, and CloudWatch metrics.

22. MODULE DESCRIPTIONS (22-26)

Authentication & Security: Admin/staff login with salted bcrypt hashing and JWT tokens.

Executive Dashboard: Overview of waste mass, active trucks, fuel, and recent activity.

Complaint Management: Citizen complaint submission with photo preview and status progression.

Fleet & Collections: Registration of compactor/tipper trucks and scheduling pickups by waste type.

GIS Map & Locations: Interactive Leaflet map showing municipal bins and fill states.

Reports & Analytics: Interactive pie, bar, and line charts of municipal performance.

27. AUTHENTICATION & 28. AWS CLOUD SERVICES

27. Security Implementation

Bcrypt Hashing: Passwords encrypted with 12 salt rounds.

JWT Tokens: HMAC-SHA256 signed bearer tokens with 8-hour expiration.

Magic-Byte Validation: File signatures checked before storage to reject malicious uploads.

SQL Defense: 100% of queries use parameterized SQL statements.

28. AWS Cloud Services Status

All AWS resources are 100% codified in aws/cloudformation.yaml and deployable via aws/deploy.ps1. Actual cloud deployment remains pending due to institutional AWS Academy lab access being unavailable during testing.

29. TESTING AND RESULTS

29. Automated Integration Test Suite (24/24 PASS)

I executed backend/test_suite.js against the running system. All 24 test cases passed successfully (100% pass rate):

Endpoints Verified: GET /api/health (200), GET /api/db-check (200), POST /api/users/login (200/401), GET/POST /api/complaints (200/201), PATCH /api/complaints/:id/status (200), GET/POST /api/vehicles (200/201), PUT /api/vehicles/:id (200), GET/POST /api/waste-collections (200/201), PUT /api/waste-collections/:id (200), GET/POST /api/collection-locations (200/201), GET /api/reports/summary (200), GET /api/users (200), Teardown cleanup (200).

30. LIMITATIONS, 31. FUTURE SCOPE & 32. CONCLUSION

30. Limitations

AWS deployment was verified through CloudFormation template validation; actual cloud provisioning is pending institutional AWS Academy access. The full prototype was verified locally and over a secure edge tunnel.

31. Future Enhancements

IoT Ultrasonic Sensors: Communal bins equipped with ESP32 sensors via AWS IoT Core.

AI Waste Sorting: Amazon Rekognition for automated waste classification from photos.

Route Optimization: Dijkstra shortest-path routing for compactor trucks.

32. Conclusion & 33. References

I successfully designed, built, and verified GreenCity as an individual mini-project. The system demonstrates enterprise cloud architecture, responsive web engineering, and robust security, satisfying all academic criteria.

References: UN SDG 11 & 12 (2023); AWS Well-Architected Framework (2024); MySQL 8.0 Reference Manual (2024); React 19 & Express.js Docs (2024).